

# Boran Yesilyurt

# Curriculum Vitae

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## Contact

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## Education

### University of Florida (UF)

Gainesville, USA

*Physics, PhD*

*August 2022 – Expected August 2027*

- Current GPA: 4.00/4.00
- Advisor: Richard P. Woodard
- Teaching Assistant for more than 8 courses
- Referee at Classical and Quantum Gravity (IOP)

### Middle East Technical University (METU)

Ankara, Turkey

*Physics, BSc*

*August 2018 – June 2022*

- Valedictorian-GPA: 4.00/4.00
- Senior Project: *Schwinger Process of Pair Production on Flat Two-Torus*
- Advisor: Seckin Kurkcuoglu

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## Skills

**Programming:** Python, Mathematica, LaTeX, JavaScript(Vanilla), HTML, CSS

**Scientific Computing:** NumPy, SciPy, Matplotlib, Sklearn, Statmodels, Lifelines, Sksurv, Symbolic computation, Pattern recognition, Numerical methods

**AI & Machine Learning:** LlamaIndex, FAISS, OpenAI API, Retrieval-Augmented Generation (RAG), Agentic AI Systems, Prompt Engineering, Tool Calling

**Quantitative Methods:** Forecasting, Time Series analysis, Stochastic modeling, Survival analysis, Machine Learning, Monte Carlo simulation, Perturbation theory, Statistical mechanics

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## Research Experience

### University of Florida, Physics Department

Advisor: Richard P. Woodard

*Research Assistantship & Doctoral Studies*

*December 2022 – Current*

- Conducting research in quantum field theory, cosmology, and quantum gravity with emphasis on infrared effects and stochastic phenomena in de Sitter spacetime.
- Independently completed a technically demanding graviton self-energy calculation involving nontrivial symbolic manipulation, pattern recognition and consistency verification across multiple analytical regimes.
- Performed symbolic and numerical computations in Python to validate theoretical predictions and study emergent large-scale stochastic behavior.
- Co-authored six peer-reviewed publications and one first-author manuscript currently in preparation.

### Middle East Technical University, Physics Department

Senior Project Advisor: Seckin Kurkcuoglu

*Independent Research and Senior Project*

*2018 – 2022*

- Studied Schwinger pair production on compact manifolds using analytical quantum field theoretic methods.
- Derived and numerically analyzed equations governing particle creation under nontrivial topological boundary conditions.
- Developed a generalized equations-of-motion framework for multi-body pendulum systems.
- Formulated and solved coupled nonlinear dynamical equations for constrained classical systems.

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## **Publications**

*Order of the authors in HEP-TH follows last name alphabetical order. Exceptions are listed with an Asterisk.*

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|----------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Yesilyurt, Tsamis, Woodard - Dimensionally Regulated Graviton Self-Energy from Gravitons on $dS^*$ | Manuscript in Preparation                     |
| Woodard, Yesilyurt - The Other ADM                                                                 | <i>J.Phys.A</i> 58 (2025) 19, 195401          |
| Miao, Tsamis, Woodard, Yesilyurt - The Third Structure Function                                    | <i>Class.Quant.Grav.</i> 42 (2025) 10, 105001 |
| Tsamis, Woodard, Yesilyurt - The Need to Renormalize Cosmological Constant                         | <i>Phys.Lett.B</i> 849 (2024) 138472          |
| Litos, Woodard, Yesilyurt - Large Logs in a Nontrivial Nonlinear Sigma Model                       | <i>Phys.Rev.D</i> 108 (2023) 6                |
| Woodard, Yesilyurt - Remembrance of Things Past                                                    | <i>JHEP</i> 08 (2023) 124                     |
| Woodard, Yesilyurt - Unfinished Business in a Nonlinear Sigma Model on $dS$ Background             | <i>JHEP</i> 06 (2023) 206                     |
| Yesilyurt - Equations of Motion of A Pendulum Containing $N$ -point Masses                         | <i>arXiv:1910.12610</i> (2019)                |

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## **Projects**

### **Return Non-Gaussianity as a Cross-Sectional Signal**

*Python, Pandas, NumPy, SciPy*

- Constructed and backtested raw skewness and coskewness signals on a 20-year, survivorship-bias-corrected daily return panel of US equities, evaluating cross-sectional predictability via portfolio sorts, Information Coefficient analysis, and Fama-MacBeth regression; found statistically significant nonzero skewness amplitude that is nonetheless unpriced, consistent with the lottery-demand literature
- Designed a model-independent decomposition, motivated by an analogy to quantum field theory, to test whether observed return non-Gaussianity is a fundamental feature of returns or an emergent artifact of time-varying volatility
- Fit per-stock GARCH(1,1)-Gaussian null models and used Monte Carlo simulation to show that volatility clustering alone explains under 0.1% of observed cross-sectional skewness amplitude, isolating a statistically significant residual unexplained by the null and tracing a structural sign reversal coinciding with the 2011 Eurozone crisis
- Identified that cross-sectional correlation between raw skewness and coskewness functions as a leading indicator of systemic stress, peaking ahead of the 2008, 2011, and 2020 crises, while maintaining explicit epistemic discipline around subtraction-scheme dependence throughout

### **Enrollment Forecasting and Survival Analysis**

*Python, Pandas, Statsmodels, Scikit-Learn, Lifelines*

- Performed end-to-end analysis of a 15-year enrollment dataset, including data cleaning, feature engineering, exploratory data analysis, and statistical visualization
- Investigated demographic and occupational effects on annual enrollment through correlation analysis, identifying occupation-specific patterns associated with long-term cohort dynamics
- Implemented and compared multiple forecasting approaches including exponential smoothing, ARIMA, Kalman filtering, Bayesian trend models, and nonlinear Fourier regression using residual diagnostics and out-of-sample validation
- Developed a custom Fourier + trend model that achieved the most reliable long-term enrollment forecasts while maintaining model simplicity and well-behaved residuals
- Applied Kaplan–Meier estimation, Random Survival Forests, and DeepSurv neural networks to model customer churn, demonstrating that static demographic variables provide limited predictive power for membership retention

### **Stochastic Modeling of Bacterial Colony Growth**

*Python, stochastic simulation, data analysis*

- Developed a phenomenological model for effective colony-size growth using symmetry-based arguments, stochastic dynamics and local interaction dynamics
- Implementing biased random-walk and Monte Carlo methods to study emergent growth behavior and compare model predictions across parameter regimes by simulating multiple independent realizations
- Computed statistical observables across runs, including mean growth profiles and variance evolution over time
- Analyzed how stochastic fluctuations affect macroscopic colony structure formation
- Built visualization tools to track spatial configuration and temporal evolution of colony states across iterations
- Building a reproducible research notebook with structured visualizations and documented future directions

### **Paper Debate Arena: Agentic Argument Mapping System**

*Python, React, FastAPI, LlamaIndex, FAISS, OpenAI*

- Built an end-to-end pipeline that ingests two research papers, detects and classifies conflicts into empirical, assumption, definitional, and methodological types, and maps them as an interactive argument graph

- Implemented per-paper RAG architecture using LlamaIndex and FAISS with section-aware chunking and source identity preservation across all retrieval and reasoning steps
- Designed three cross-examination tools that ground every agent response in retrieved passages with section and page references
- Developed a FastAPI backend with session persistence and a React frontend with React Flow graph visualization, supporting export and reload of argument maps across sessions

#### **Interactive Elden Ring Companion App**

*Python, JavaScript, HTML, CSS*

- Designed a companion utility tool for organizing and accessing structured game information providing an interactive game data exploration experience
- Implemented dynamic UI logic and state management using Vanilla JavaScript
- Built modular architecture to allow easy extension of content categories and future updates
- Improved user experience by centralizing game information and providing in-session AI assistance

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## **Scholarships & Awards**

|                                                               |                                  |
|---------------------------------------------------------------|----------------------------------|
| <i>Institute of Fundamental Theory(IFT) Summer Fellowship</i> | <i>May - August 2025, UF</i>     |
| <i>DGRAV APS 2025 Travel Grant</i>                            | <i>2025, Anaheim,CA</i>          |
| <i>Charles F. Hooper Jr. Memorial Award</i>                   | <i>2024, UF</i>                  |
| <i>TRISEP 2023 Travel Grant</i>                               | <i>2023, Perimeter Institute</i> |
| <i>Teaching Fellowship</i>                                    | <i>2022-Current, UF</i>          |
| <i>TÜBİTAK - 2205 Scholarship</i>                             | <i>2017-2022, METU</i>           |
| <i>Yükseköğretim Kurulu (YÖK) Destek Scholarship</i>          | <i>2021-2022, METU</i>           |

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## **Talks**

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|-----------------------------------------------------------------------------|---------------------------------------------------|
| <i>2025 APS Joint April &amp; March Meeting - Poster Presentation</i>       | <i>16-21 March, 2025, Anaheim,CA</i>              |
| <i>10th Gulf Coast Gravity Meeting - Talk</i>                               | <i>1-2 March, 2024, UT-Austin,TX</i>              |
| <i>The Early Universe: A Window to New Physics - Talk</i>                   | <i>20-22 October, 2023, University of Florida</i> |
| <i>TRISEP 2023 Summer School, Perimeter Institute - Poster Presentation</i> | <i>19-30 June, 2023, Ontario/Canada</i>           |

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## **Teaching**

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|--------------------------------------------------------------|------------------------------|
| <i>Physics - 1: Labs</i>                                     | <i>University of Florida</i> |
| <i>Physics - 1: Discussion Teaching Assistant(TA)</i>        | <i>University of Florida</i> |
| <i>PHZ3113 Intro to Theoretical Physics: TA</i>              | <i>University of Florida</i> |
| <i>PHY4324 Electromagnetism 2: TA</i>                        | <i>University of Florida</i> |
| <i>PHY4604 Quantum Mechanics 1: TA</i>                       | <i>University of Florida</i> |
| <i>PHY6347 (Graduate Level) Electromagnetic Theory 2: TA</i> | <i>University of Florida</i> |
| <i>PHY6646 (Graduate Level) Quantum Mechanics 2: TA</i>      | <i>University of Florida</i> |

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## **Service, Outreach & Event Organization**

### **Referee**

*Classical and Quantum Gravity (IOP)*

### **Graduate Students and Postdoc Seminars (GSPS)**

*UF*

*Monthly HEP-TH Seminar Organizer*

*September 2023 – Ongoing*

### **The 62nd Florida Regional Junior Science and Humanities Symposium**

*UF*

*Referee for Admission to STEM Research Speaker Competition*

*2-4 February, 2025*

### **Physics Society**

*METU*

*Board Member between March 2018-2019 September*

*October 2017 – October 2019*

- Demonstrated experiments in Yenimahalle Science Fair- 2019, Ankara
- Organized Interscholastic Physics Students Summit (IPSS)- 2018 and IPSS- 2019
- Coordinated and organized weekly seminars with lecturers from Physics department on topics out of the curriculum
- Attended International Conference of Physics Students (ICPS)- 2019 held at Cologne, Germany as representative and participant

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## REFERENCES

- Available upon request.